

Lecture 8: Basic New Keynesian

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May 2026

RBC to NK

RBC: Fluctuations explained by technology shocks (kydland–Prescott 1982; Prescott 1986)

Use DSGE models as tool: FOCs replacing behavioral equations, rational expectations.

1. Efficient allocation.
2. Implication: stabilization policies non-necessary.
3. Equilibrium outcome responds to exogenous shocks.
4. Limited role of monetary factors, neutrality of monetary policy: Friedman's rule

* Contradiction to:

1. Common belief: influence on output and employment.
2. Empirical challenge: non-neutrality.

⇒ Irrelevancy to policy evaluation: Normative implication different from policy practice.

Monetary policy can affect real variables or not: e.g. output, employment, consumption.

RBC

Agents / household / firms

DSGE

Markets are frictionless and perfectly competitive

Price adjust instantaneously

Technology shocks

Not really: RBC also can introduce other shocks.

New Keynesian

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Goods market imperfection (monopolistic competition)

Nominal rigidity (price, wage etc.)

Other shocks and transmission mechanisms

New in NK

1. Monopolistic competition: $P^m > P^c, Y^m < Y^c$, 垄断竞争 inefficient response to shocks.
2. Nominal rigidities: staggered price / wage, 名义刚性, 价格短期刚性.
3. Short-run non-neutrality of monetary policy, 短期货币非中性 $\Rightarrow$ policy analysis & intervention.

These elements are new to RBC but old to Keynesian (static, reduced-form).

Empirical evidence of nominal rigidities:

- Taylor (1999): Average price adjustment frequency = 1 year.
- Bils and Klenow (2004): Heterogeneity of frequency across sectors / types of goods.
- Nakamura and Steinsson (2008): Adjusting for sales, frequency $\approx$ 8–11 months.
- Bewley (1998): Wages do not fall during recessions (downward nominal rigidity).

* High welfare / social security countries face stronger downward wage rigidity.

? Working hour rigidity in China.

$$\text{Nominal rigidity} \Rightarrow \text{price inertia} \Rightarrow \begin{cases} \text{Monetary policy changes real balance } M/P, \\ \text{change real interest rate } r = i - \pi. \end{cases}$$

$\Rightarrow$ Change in aggregate demand $\Rightarrow$ Change in employment/output.

Identify monetary policy shock:

$$\begin{cases} \text{endogenous movement:} & \text{deliberate response to developments in economy} \Rightarrow \text{co-movement,} \\ \text{exogenous movement:} & \text{structural VAR residuals from policy rule.} \end{cases}$$

$$\begin{cases} \text{endogenous comovements:} & \begin{aligned} & \text{high interest rate, high inflation/high output,} \\ & i_t = \phi_\pi \pi_t + \phi_y y_t + v_t, \\ & \text{policy response + exogenous component / unexpected shocks;} \end{aligned} \\ \text{exogenous component:} & u_t \text{ identification requires: GDP and prices cannot respond} \\ & \text{contemporaneously to a monetary policy shock.} \end{cases}$$

Basic NK

Basic setup:

- ① DSGE structure: Households live infinitely, maximize utility subject to intertemporal budget constraint. Firms with identical technology, subject to random shocks.
- ② AS and AD block.

AD: log-linearized consumption Euler (relates output and price).

Let $\hat{x}_t = \log X_t - \log \bar{X}$ except i_t, r_t, π_t .

$$\tilde{y}_t = \hat{y}_t - \hat{y}_t^n$$

$\hat{y}_t^n$: natural output under flexible prices / long-run output.

$$\tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\sigma} \underbrace{\left(\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1} \right)}_{\hat{r}_t} + \underbrace{v_t}_{\text{MP shock}} . \quad (1)$$

AS: New Keynesian Phillips Curve (relates price and output):

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa \tilde{y}_t. \quad (2)$$

- ③ Policy rule:

$$\hat{i}_t = f(\Omega_t) + v_t, \quad \Omega_t : \text{information set / state vector}. \quad (3)$$

Four markets: goods, labor, bonds, money (cashless).

Differentiated goods: continuum of varieties $i \in [0, 1]$.

Perfect competition in labor market, no wage rigidities.

Monopolistic competition on goods market: goods prices set by firms that produce them (profit $\neq 0$).

Nominal prices are sticky: a fraction $(1 - \theta)$ of firms can reset their prices each period (Calvo pricing).

A continuum of infinitely-lived households.

A continuum of firms represented by $[0, 1]$ goods $\Rightarrow$ 1 firm produces 1 differentiated good and sets price.

(i) Household's problem

$$\begin{aligned} & \max_{C_t, N_t, B_t} \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t U(C_t, N_t), \\ \text{s.t. } & \boxed{P_t} C_t + Q_t B_t = B_{t-1} + W_t N_t + T_t. \end{aligned}$$

P_t : aggregate price index of the consumption bundle, $P_t C_t = \int_0^1 P_t(i) C_t(i) di$.

$$C_t = \left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}},$$

C_t : A composite bundle of goods; continuum of varieties produced by continuum of firms.

i : variety (differentiated goods) index,

di : continuum of varieties produced by continuum of firms,

ε : across-variety EoS.

Solve household's problem in two steps:

1. Static optimal expenditure allocation (how to allocate across varieties?)

$$E = \min_{C_t(i)} \int_0^1 P_t(i) C_t(i) di = \text{total expenditure} \quad \text{s.t.} \quad C_t = \left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}}.$$

$$\mathcal{L} = \int_0^1 P_t(i) C_t(i) di - P_t \left[\left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}} - C_t \right].$$

$$\frac{\partial \mathcal{L}}{\partial C_t(i)} = 0 \Rightarrow P_t(i) - P_t \frac{\varepsilon}{\varepsilon-1} \left[\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di \right]^{\frac{1}{\varepsilon-1}} \frac{\varepsilon-1}{\varepsilon} C_t(i)^{-\frac{1}{\varepsilon}} = 0$$

$$\Rightarrow P_t(i) - P_t C_t^{1/\varepsilon} C_t(i)^{-1/\varepsilon} = 0$$

$$\Rightarrow C_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon} C_t.$$

$$C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} = \left(\frac{P_t(i)}{P_t} \right)^{1-\varepsilon} C_t^{\frac{\varepsilon-1}{\varepsilon}},$$

$$\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di = C_t^{\frac{\varepsilon-1}{\varepsilon}} \int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{1-\varepsilon} di,$$

$$C_t^{\frac{\varepsilon-1}{\varepsilon}} = C_t^{\frac{\varepsilon-1}{\varepsilon}} \int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{1-\varepsilon} di,$$

$$P_t^{1-\varepsilon} = \int_0^1 P_t(i)^{1-\varepsilon} di, \quad P_t = \left(\int_0^1 P_t(i)^{1-\varepsilon} di \right)^{\frac{1}{1-\varepsilon}}.$$

Why ε is the elasticity of substitution across varieties?

$$C_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon} C_t, \quad C_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\varepsilon} C_t \Rightarrow \frac{C_t(i)}{C_t(j)} = \left(\frac{P_t(i)}{P_t(j)} \right)^{-\varepsilon}.$$

$$\log \frac{C_t(i)}{C_t(j)} = -\varepsilon \log \frac{P_t(i)}{P_t(j)}, \quad \frac{d \log \{C_t(i)/C_t(j)\}}{d \log \{P_t(i)/P_t(j)\}} = -\varepsilon.$$

$$\sigma_{ij} = -\frac{d \log \{C_t(i)/C_t(j)\}}{d \log \{P_t(i)/P_t(j)\}} = \varepsilon, \quad \text{elasticity of substitution between two varieties.}$$

2. Intertemporal problem

$$\max_{C_t, N_t, B_t} \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t U(C_t, N_t) \quad \text{s.t.} \quad P_t C_t + Q_t B_t = B_{t-1} + W_t N_t + T_t.$$

$$\mathcal{L} = \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t \{U(C_t, N_t) + \lambda_t (B_{t-1} + W_t N_t + T_t - P_t C_t - Q_t B_t)\}.$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \Rightarrow U_{C,t} = P_t \lambda_t \quad (1),$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = 0 \Rightarrow U_{N,t} + \lambda_t W_t = 0 \quad (2),$$

$$\frac{\partial \mathcal{L}}{\partial B_t} = 0 \Rightarrow \beta^{t+1} \mathbb{E}_t \lambda_{t+1} - \beta^t \lambda_t Q_t = 0 \Rightarrow \mathbb{E}_t \beta \lambda_{t+1} = \lambda_t Q_t \quad (3).$$

$$(1) \& (2) \Rightarrow -\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t} \quad \text{Labor-leisure condition (intratemporal).}$$

$$(1) \& (3) \Rightarrow Q_t = \beta \mathbb{E}_t \left\{ \frac{U_{C,t+1}}{U_{C,t}} \frac{P_t}{P_{t+1}} \right\} = \beta \mathbb{E}_t \left\{ \frac{U_{C,t+1}}{U_{C,t}} \frac{1}{\Pi_{t+1}} \right\}, \quad \Pi_{t+1} = \frac{P_{t+1}}{P_t} \text{ inflation.}$$

Bond Euler equation (intertemporal).

Suppose we have a separable utility function

$$U(C_t, N_t) = \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi}, \quad U_{C,t} = C_t^{-\sigma}, \quad U_{N,t} = -N_t^{\varphi}.$$

$$\frac{W_t}{P_t} = N_t^{\varphi} C_t^{\sigma}, \quad Q_t = \beta \mathbb{E}_t \left\{ \frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}} \frac{1}{\Pi_{t+1}} \right\}.$$

Log-linearize:

$$\hat{w}_t - \hat{p}_t = \sigma \hat{c}_t + \varphi \hat{n}_t, \quad \hat{c}_t = \mathbb{E}_t \hat{c}_{t+1} - \frac{1}{\sigma} (\hat{i}_t - \hat{\Pi}_{t+1}).$$

$$Q_t = e^{-i_t}, \quad \hat{\Pi}_{t+1} = \log \Pi_{t+1} - \log \bar{\Pi} = \log \Pi_{t+1} = \pi_{t+1}.$$

(ii) Firm's problem: two simultaneous problems (isomorphic)

1. Cost minimization for each producer in the continuum

$$\min_{N_t(i)} W_t N_t(i) \quad \text{s.t.} \quad Y_t(i) = A_t N_t(i)^{1-\alpha}.$$

$$\mathcal{L} = W_t N_t(i) + \mu_t [Y_t(i) - A_t N_t(i)^{1-\alpha}].$$

$$\frac{\partial \mathcal{L}}{\partial N_t(i)} = W_t - \mu_t (1-\alpha) A_t N_t(i)^{-\alpha} = 0 \Rightarrow W_t = \mu_t (1-\alpha) A_t N_t(i)^{-\alpha}.$$

μ_t is the Lagrangian multiplier: shadow value of relaxing the production requirement by one unit.

$\Rightarrow \mu_t$ is the nominal marginal cost: $\mu_t = MC_t$, $\mu_t = MC_t = \frac{W_t}{MPL_t(i)}$.

W_t : 1 labor's cost; $MPL_t(i)$: 1 labor's output; $\frac{W_t}{MPL_t(i)}$: cost per unit of output.

2. Profit maximization by choosing price.

$$\max_{P_t(i)} P_t(i)Y_t(i) - W_tN_t(i) = \max_{P_t(i)} P_t(i)Y_t(i) - W_t \left(\frac{Y_t(i)}{A_t} \right)^{\frac{1}{1-\alpha}} \quad \text{production function}$$

$$\frac{\partial Profit}{\partial P_t(i)} = Y_t(i) + P_t(i) \frac{\partial Y_t(i)}{\partial P_t(i)} - W_t \frac{1}{1-\alpha} \left(\frac{Y_t(i)}{A_t} \right)^{\frac{\alpha}{1-\alpha}} \frac{\partial Y_t(i)}{\partial P_t(i)} \frac{1}{A_t} = 0. \quad (*)$$

$$MC_t = \frac{\partial}{\partial Y_t(i)} \left[W_t \left(\frac{Y_t(i)}{A_t} \right)^{\frac{1}{1-\alpha}} \right] = W_t \frac{1}{1-\alpha} \left(\frac{Y_t(i)}{A_t} \right)^{\frac{\alpha}{1-\alpha}} \frac{1}{A_t}$$

$$Y_t(i) + [P_t(i) - MC_t] \frac{\partial Y_t(i)}{\partial P_t(i)} = 0.$$

Plug in:

$$Y_t(i) - \varepsilon [P_t(i) - MC_t] \frac{Y_t(i)}{P_t(i)} = 0.$$

Divide both sides by $Y_t(i)$:

$$1 - \varepsilon \frac{P_t(i) - MC_t}{P_t(i)} = 0.$$

$$1 - \varepsilon + \varepsilon \frac{MC_t}{P_t(i)} = 0.$$

Therefore,

$$P_t(i) = \frac{\varepsilon}{\varepsilon - 1} MC_t.$$

Variety price equals to a constant markup μ over the marginal cost

$$\text{where } \mu = \frac{\varepsilon}{\varepsilon - 1}; \quad \text{monopolistic competition pricing.}$$

When $\varepsilon \rightarrow \infty \Rightarrow \mu \rightarrow 1$, perfect competition, $P_t(i) = MC_t$.

In general ε is finite, $\mu > 1$, $P_t(i) > MC_t \Rightarrow$ inefficiency, low employment and output.

Cost minimization 1 and profit maximization 2 are dual problems:

1. tells how expensive it is to produce at each output level.
2. tells the price-taking that cost schedule and demand as given.

Nesting the two:

$$C(Y_t(i)) = \min_{N_t(i)} W_t N_t(i) \quad \text{s.t.} \quad Y_t(i) = A_t N_t(i)^{1-\alpha}.$$

Then solve

$$\max_{P_t(i)} \{P_t(i)Y_t(i) - C(Y_t(i))\} \quad \text{s.t.} \quad Y_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon} C_t.$$

When $\alpha = 0$, marginal cost connects the two problems:

$$MC_t = \frac{\partial C(Y_t(i))}{\partial Y_t(i)}.$$

(iii) Sticky-price equilibrium

Calvo (1983) pricing: Each period a measure $(1 - \theta)$ fraction of the continuum of firms reset their prices, the other θ keeps price unchanged.

$$\Rightarrow \text{Average duration of a price is } \frac{1}{1 - \theta}.$$

θ : index of price stickiness, pinned down by data.

Thus

$$P_t = [\theta P_{t-1}^{1-\varepsilon} + (1 - \theta)(P_t^*)^{1-\varepsilon}]^{\frac{1}{1-\varepsilon}}.$$

Log-linearize:

$$\begin{aligned} P_t^{1-\varepsilon} &= \theta P_{t-1}^{1-\varepsilon} + (1 - \theta)(P_t^*)^{1-\varepsilon}, \\ \bar{P}^{1-\varepsilon}(1 + \hat{p}_t)^{1-\varepsilon} &= \theta \bar{P}^{1-\varepsilon}(1 + \hat{p}_{t-1})^{1-\varepsilon} + (1 - \theta)\bar{P}^{1-\varepsilon}(1 + \hat{p}_t^*)^{1-\varepsilon}, \\ \bar{P}^{1-\varepsilon}(1 - \varepsilon)\hat{p}_t &= \theta \bar{P}^{1-\varepsilon}(1 - \varepsilon)\hat{p}_{t-1} + (1 - \theta)\bar{P}^{1-\varepsilon}(1 - \varepsilon)\hat{p}_t^*, \\ \hat{p}_t &= \theta \hat{p}_{t-1} + (1 - \theta)\hat{p}_t^*, \\ \hat{p}_t - \hat{p}_{t-1} &= (1 - \theta)(\hat{p}_t^* - \hat{p}_{t-1}). \end{aligned}$$

$$\begin{aligned} \hat{\pi}_t &= \pi_t - \bar{\pi} = \pi_t = \log P_t - \log P_{t-1} = \log P_t - \log \bar{P} - (\log P_{t-1} - \log \bar{P}) \\ &= \hat{p}_t - \hat{p}_{t-1}. \end{aligned}$$

$$\Rightarrow \hat{\pi}_t = (1 - \theta)(\hat{p}_t^* - \hat{p}_{t-1}).$$

* Positive inflation arises iff firms that adjust prices in t choose to charge above the price level prevailed in $t - 1$.

$\theta \uparrow \Rightarrow$ more sticky firms $\Rightarrow$ inflation is lower given the price difference.

Due to sticky price, firms must set price taking into account the risk that they will not be able to change price in the future.

$$\begin{aligned} V_t(i) &= P_t(i)Y_t(i) - cost_t(i) + \theta \mathbb{E}_t\{Q_{t,t+1}[P_t(i)Y_{t+1}(i) - cost_{t+1}(i)]\} \\ &\quad + \theta^2 \mathbb{E}_t\{Q_{t,t+2}[P_t(i)Y_{t+2}(i) - cost_{t+2}(i)]\} + \dots \end{aligned}$$

Optimal pricing under sticky price: Choose P_t^* to maximize the expected PV of profits while this price remains effective, $P_t^* = P_t^*(i)$.

$$V_t(i) = \max_{P_t^*} \sum_{k=0}^{\infty} \theta^k \mathbb{E}_t [Q_{t,t+k}(P_t^* Y_{t+k|t} - \psi_{t+k}(Y_{t+k|t}))] \quad \text{conditional on the firm can reset at } t.$$

$$\text{s.t. } Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}} \right)^{-\varepsilon} C_{t+k}.$$

$$Q_{t,t+k} = \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \left(\frac{P_t}{P_{t+k}} \right) \quad \text{stochastic discount factor for nominal payoffs.}$$

$\psi_{t+k}(\cdot)$ is the cost function.

$Y_{t+k|t}$ is output in period $t+k$ for a firm that last reset price at t .

When $\theta = 0$, collapse to with only $k = 0$ term $\Rightarrow P_t^* = \frac{\varepsilon}{\varepsilon-1} \psi_t$.

* $1 - \theta$ determines how many firms reset in the cross section.

θ^k matters for how long a chosen reset price survives in the time dimension.

FOC:

$$\begin{aligned} \frac{\partial V_t(i)}{\partial P_t^*} = 0 &\Rightarrow \mathbb{E}_t \left\{ \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} \left[Y_{t+k|t} + P_t^* \frac{\partial Y_{t+k|t}}{\partial P_t^*} - \frac{\partial \psi_{t+k}}{\partial Y_{t+k|t}} \frac{\partial Y_{t+k|t}}{\partial P_t^*} \right] \right\} = 0, \\ \frac{\partial Y_{t+k|t}}{\partial P_t^*} &= -\varepsilon \left(\frac{P_t^*}{P_{t+k}} \right)^{-\varepsilon-1} \frac{C_{t+k}}{P_{t+k}} = -\varepsilon \frac{Y_{t+k|t}}{P_t^*}. \end{aligned}$$

Define

$$\begin{aligned} \varphi_{t+k|t} &= \frac{\partial \psi_{t+k}}{\partial Y_{t+k|t}} \quad \text{as nominal marginal cost.} \\ \mathbb{E}_t \left\{ \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} \left[Y_{t+k|t} + (P_t^* - \varphi_{t+k|t})(-\varepsilon) \frac{Y_{t+k|t}}{P_t^*} \right] \right\} &= 0, \\ \mathbb{E}_t \left\{ \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} Y_{t+k|t} \left[1 - \varepsilon + \varepsilon \frac{\varphi_{t+k|t}}{P_t^*} \right] \right\} &= 0. \\ P_t^* &= \underbrace{\frac{\varepsilon}{\varepsilon-1}}_{\text{markup } \mu} \cdot \underbrace{\frac{\mathbb{E}_t \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} Y_{t+k|t} \varphi_{t+k|t}}{\mathbb{E}_t \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} Y_{t+k|t}}}_{\text{weighted}} \end{aligned}$$

Weighted average of future marginal cost(*).

$$\frac{\theta^k Q_{t,t+k} Y_{t+k|t}}{\mathbb{E}_t \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} Y_{t+k|t}} : \text{weight/how nominal MC of } t+k \text{ matters.}$$

When $\theta \rightarrow 0$, $P_t^* \rightarrow \frac{\varepsilon}{\varepsilon-1} \varphi_t$.

Weights are larger when:

- 1) $\theta \uparrow$: price is more sticky;
- 2) $Q_{t,t+k} \uparrow$: future profit is more valuable in utility terms;
- 3) $Y_{t+k|t} \uparrow$: firms expect to sell more in that period.

Why $Q_{t,t+k} = \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \left(\frac{P_t}{P_{t+k}} \right)$?

$$Q_t = \beta \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} \right] \quad \text{one-period nominal discount factor.}$$

$$\begin{aligned}
Q_t &= Q_{t,t+1}, \\
Q_{t+1} &= \beta \left[\left(\frac{C_{t+2}}{C_{t+1}} \right)^{-\sigma} \frac{P_{t+1}}{P_{t+2}} \right], \\
Q_{t,t+2} &= Q_{t,t+1} Q_{t+1,t+2} = \beta^2 \left(\frac{C_{t+2}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+2}}, \\
Q_{t,t+k} &= \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+k}}.
\end{aligned}$$

β^k : impatience.

$\left(\frac{C_{t+k}}{C_t} \right)^{-\sigma}$: marginal utility of future consumption.

$\frac{P_t}{P_{t+k}}$: inflation erosion of nominal payoff; converts nominal payoff into real payoff.

$$Q_{t,t+k} = \beta^k \tilde{Q}_{t,t+k}, \quad \tilde{Q}_{t,t+k} = \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma} \left(\frac{P_t}{P_{t+k}} \right).$$

$$Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}} \right)^{-\varepsilon} C_{t+k} = (P_t^*)^{-\varepsilon} P_{t+k}^\varepsilon Y_{t+k}, \quad (Y_t = C_t, \forall t).$$

$$(*) \Rightarrow P_t^* = \mu \frac{\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \tilde{Q}_{t,t+k} P_{t+k}^\varepsilon Y_{t+k} \varphi_{t+k|t}}{\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \tilde{Q}_{t,t+k} P_{t+k}^\varepsilon Y_{t+k}}.$$

$$P_t^* \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \tilde{Q}_{t,t+k} P_{t+k}^\varepsilon Y_{t+k} = \mu \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \tilde{Q}_{t,t+k} P_{t+k}^\varepsilon Y_{t+k} \varphi_{t+k|t}.$$

Let

$$A_t \equiv \sum_{k=0}^{\infty} (\beta\theta)^k \tilde{Q}_{t,t+k} P_{t+k}^\varepsilon Y_{t+k}, \quad a_{t+k} \equiv (\beta\theta)^k \tilde{Q}_{t,t+k} P_{t+k}^\varepsilon Y_{t+k}.$$

$$\widehat{LHS} = \hat{p}_t^* + \mathbb{E}_t \hat{A}_t.$$

$$\hat{A}_t = \sum_{k=0}^{\infty} \frac{\bar{a}_{t+k}}{\bar{A}} \hat{a}_{t+k},$$

$$\frac{\bar{a}_{t+k}}{\bar{A}} = \frac{(\beta\theta)^k \tilde{Q} \bar{P}^\varepsilon \bar{Y}}{\sum_{k=0}^{\infty} (\beta\theta)^k \tilde{Q} \bar{P}^\varepsilon \bar{Y}} = \frac{(\beta\theta)^k \tilde{Q} \bar{P}^\varepsilon \bar{Y}}{\frac{1}{1-\beta\theta} \tilde{Q} \bar{P}^\varepsilon \bar{Y}} = (1-\beta\theta)(\beta\theta)^k,$$

$$\hat{a}_{t+k} = \hat{q}_{t,t+k} + \varepsilon \hat{p}_{t+k} + \hat{y}_{t+k}.$$

$$\widehat{LHS} = \hat{p}_t^* + (1-\beta\theta) \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k (\hat{q}_{t,t+k} + \varepsilon \hat{p}_{t+k} + \hat{y}_{t+k}).$$

$$B_t \equiv \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \tilde{Q}_{t,t+k} P_{t+k}^\varepsilon Y_{t+k} \varphi_{t+k|t}, \quad \widehat{RHS} = \mathbb{E}_t \hat{B}_t.$$

$$\hat{B}_t = \sum_{k=0}^{\infty} \frac{\bar{b}_{t+k}}{\bar{B}} \hat{b}_{t+k}, \quad \frac{\bar{b}_{t+k}}{\bar{B}} = (1-\beta\theta)(\beta\theta)^k,$$

$$\hat{b}_{t+k} = \hat{q}_{t,t+k} + \varepsilon \hat{p}_{t+k} + \hat{y}_{t+k} + \hat{\varphi}_{t+k|t}.$$

$$\widehat{RHS} = (1 - \beta\theta)\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k (\hat{q}_{t,t+k} + \varepsilon \hat{p}_{t+k} + \hat{y}_{t+k} + \hat{\varphi}_{t+k|t}).$$

Combine $\widehat{LHS} = \widehat{RHS}$:

$$\begin{aligned} \hat{p}_t^* &= (1 - \beta\theta)\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \hat{\varphi}_{t+k|t} \\ &= (1 - \beta\theta)\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k (\widehat{mc}_{t+k|t}^r + \hat{p}_{t+k}). \end{aligned}$$

nominal MC = real MC $\times$ price

$$\varphi_{t+k|t} = MC_{t+k|t}^r \cdot P_{t+k} \Rightarrow \hat{\psi}_{t+k|t} = \widehat{mc}_{t+k|t}^r + \hat{p}_{t+k}.$$

1. Weights $\sum_{k=0}^{\infty} (1 - \beta\theta)(\beta\theta)^k = 1$: weighted average over expected future nominal marginal cost.
2. Aggregate price level denotes willingness to maintain (in expectation) the relative price unchanged.

Suppose $\widehat{mc}_{t+k|t}^r = 0$ for all k , real marginal cost is constant.

$$\hat{p}_t^* = (1 - \beta\theta)\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \hat{p}_{t+k}.$$

* Reset price rises when expected future aggregate prices rise. Front-load future inflation to today's reset price. Expected inflation $\Rightarrow \hat{p}_t^* \uparrow$.

3. Real MC $\widehat{mc}_{t+k|t}^r$ denotes desire to change the expected relative price to avoid any gap that may emerge between expected and desired markup. Monopolistic competitive firm's desired markup rule: $P = \mu MC$.

Thus, firm wants its real price $\frac{P_t^*}{P_{t+k}}$ to be high enough relative to real MC.

$$\hat{p}_t^* - \hat{p}_{t+k} = \widehat{mc}_{t+k|t}^r + \text{desired markup} \Rightarrow \widehat{mc}_{t+k|t}^r \uparrow \Rightarrow \hat{p}_t^* - \hat{p}_{t+k} \uparrow.$$

$\Rightarrow$ Expected future real marginal cost is higher: the firm wants a higher relative price.

(iv) Equilibrium

1. Goods market clearing:

$$Y_t(i) = C_t(i), \quad \forall i \in [0, 1].$$

Define

$$Y_t = \left(\int_0^1 Y_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}}.$$

Get $Y_t = C_t$. Log-linearize: $\hat{y}_t = \hat{c}_t$. Since

$$\begin{aligned} \hat{c}_t &= \mathbb{E}_t \hat{c}_{t+1} - \frac{1}{\sigma} (\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1}), \\ \Rightarrow \hat{y}_t &= \mathbb{E}_t \hat{y}_{t+1} - \frac{1}{\sigma} (\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1}). \end{aligned}$$

2. Labor market clearing:

$$\begin{aligned} N_t &= \int_0^1 N_t(i) di. \\ Y_t(i) &= A_t N_t(i)^{1-\alpha} \Rightarrow N_t = \int_0^1 \left(\frac{Y_t(i)}{A_t} \right)^{\frac{1}{1-\alpha}} di, \\ Y_t(i) &= \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon} Y_t \Rightarrow N_t = \left(\frac{Y_t}{A_t} \right)^{\frac{1}{1-\alpha}} \int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{-\frac{\varepsilon}{1-\alpha}} di, \\ &\Rightarrow N_t^{1-\alpha} = \frac{Y_t}{A_t} \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{-\frac{\varepsilon}{1-\alpha}} di \right]^{1-\alpha}. \end{aligned}$$

Price dispersion

Let

$$D_t = \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{-\frac{\varepsilon}{1-\alpha}} di \right]^{1-\alpha} = \exp(d_t).$$

$$Y_t = A_t N_t^{1-\alpha} D_t^{-1}, \quad D_t \text{ measures how uneven prices are across firms.}$$

$$\text{When all firms charge the same price: } \frac{P_t(i)}{P_t} = 1, \forall i \Rightarrow D_t = 1 \Rightarrow d_t = 0.$$

Reduced to $Y_t = A_t N_t^{1-\alpha}$: frictionless benchmark under flexible prices, no price distortion.

But under Calvo pricing, some firms use old price, some firms use new price $\Rightarrow D_t > 1$.

$$Y_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon} Y_t.$$

Firms with low relative prices sell too much; firms with high relative prices sell too little.

$$d_t = (1 - \alpha) \log \left[\int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{-\frac{\varepsilon}{1-\alpha}} di \right].$$

Define $p_t(i) = \log P_t(i)$, $p_t = \log P_t$, $P_t(i)/P_t = e^{p_t(i) - p_t}$. Let $\chi_i = p_t(i) - p_t$.

$$\Rightarrow d_t = (1 - \alpha) \log \left[\int_0^1 \exp \left(-\frac{\varepsilon}{1 - \alpha} \chi_i \right) di \right].$$

Second-order Taylor expansion around zero-inflation / zero-dispersion steady state.

Why zero inflation $\bar{\Pi} = 1 \Leftrightarrow$ zero dispersion $P_t(i)/P_t = 1 \Leftrightarrow \bar{d} = 0$?

$$P_t = [\theta P_{t-1}^{1-\varepsilon} + (1 - \theta)(P_t^*)^{1-\varepsilon}]^{\frac{1}{1-\varepsilon}}, \quad \text{to have } P_t = P_{t-1} = \bar{P}, \text{ need } P_t^* = \bar{P}.$$

The firms that reset price set the same price as others.

$$\text{Since } e^z = e^0 + \frac{(e^z)|_{z=0}}{1!}(z - 0) + \frac{(e^z)|_{z=0}}{2!}(z - 0)^2 + o(z^2).$$

$$e^z \approx 1 + z + \frac{z^2}{2} \quad \text{for small } z.$$

$$e^{-\frac{\varepsilon}{1-\alpha}\chi_i} \approx 1 - \frac{\varepsilon}{1-\alpha}\chi_i + \frac{1}{2} \left(\frac{\varepsilon}{1-\alpha} \right)^2 \chi_i^2.$$

$$\int_0^1 e^{-\frac{\varepsilon}{1-\alpha}\chi_i} di \approx 1 - \frac{\varepsilon}{1-\alpha} \int_0^1 \chi_i di + \frac{1}{2} \left(\frac{\varepsilon}{1-\alpha} \right)^2 \int_0^1 \chi_i^2 di,$$

$$\chi_i = p_t(i) - p_t \Rightarrow \int_0^1 \chi_i di = \int_0^1 p_t(i) di - \int_0^1 p_t di = p_t - p_t = 0,$$

$$\int_0^1 e^{-\frac{\varepsilon}{1-\alpha}\chi_i} di \approx 1 + \frac{1}{2} \left(\frac{\varepsilon}{1-\alpha} \right)^2 \int_0^1 \chi_i^2 di.$$

$$\int_0^1 \chi_i^2 di \approx \text{Var}_i\{p_t(i) - p_t\} \quad \text{cross-sectional second moment.}$$

Take logs:

$$d_t = (1 - \alpha) \log \left[1 + \frac{1}{2} \left(\frac{\varepsilon}{1 - \alpha} \right)^2 \text{Var}_i\{p_t(i) - p_t\} \right],$$

since $\log(1 + u) \approx u$,

$$d_t \approx (1 - \alpha) \frac{1}{2} \left(\frac{\varepsilon}{1 - \alpha} \right)^2 \text{Var}_i\{p_t(i) - p_t\},$$

$$d_t \approx \frac{\varepsilon^2}{2(1 - \alpha)} \text{Var}_i\{p_t(i) - p_t\},$$

$$d_t \propto \text{Var}_i\{p_t(i)\}.$$

Thus, in the neighborhood of $\hat{\pi}_t = 0$ or $\bar{\Pi} = 1$, d_t is 0 up to first-order approximation:

$$\hat{d}_t \approx 0.$$

Thus

$$Y_t = A_t N_t^{1-\alpha} D_t^{-1} \quad \text{log linearization}$$

$$\hat{y}_t = \hat{a}_t + (1 - \alpha)\hat{n}_t - \hat{d}_t \Rightarrow \hat{y}_t = \hat{a}_t + (1 - \alpha)\hat{n}_t, \quad \hat{n}_t = \frac{1}{1 - \alpha}(\hat{y}_t - \hat{a}_t).$$

Average real marginal cost

Average firm:

$$\begin{aligned}
 MC_t^r &= \frac{W_t}{P_t} \frac{1}{MPN_t} = \frac{W_t}{P_t} \frac{1}{(1-\alpha)A_t N_t^{-\alpha}}. \\
 \widehat{mc}_t^r &= \hat{w}_t - \hat{p}_t - \widehat{mpn}_t = \hat{w}_t - \hat{p}_t - (\hat{a}_t - \alpha \hat{n}_t) \\
 &= \hat{w}_t - \hat{p}_t - \left(\hat{a}_t - \frac{\alpha}{1-\alpha} \hat{y}_t + \frac{\alpha}{1-\alpha} \hat{a}_t \right) \\
 &= \hat{w}_t - \hat{p}_t - \frac{1}{1-\alpha} (\hat{a}_t - \alpha \hat{y}_t).
 \end{aligned}$$

For firms that set price at t and remains at $t+k$:

$$\begin{aligned}
 \widehat{mc}_{t+k|t}^r &= \hat{w}_{t+k} - \hat{p}_{t+k} - \widehat{mpn}_{t+k|t} \\
 &= \hat{w}_{t+k} - \hat{p}_{t+k} - \frac{1}{1-\alpha} (\hat{a}_{t+k} - \alpha \hat{y}_{t+k|t}) \\
 &= \hat{w}_{t+k} - \hat{p}_{t+k} - \frac{1}{1-\alpha} (\hat{a}_{t+k} - \alpha \hat{y}_{t+k} + \alpha \hat{y}_{t+k} - \alpha \hat{y}_{t+k|t}) \\
 &= \widehat{mc}_{t+k}^r - \frac{\alpha}{1-\alpha} (\hat{y}_{t+k} - \hat{y}_{t+k|t}).
 \end{aligned}$$

Since

$$\begin{aligned}
 Y_{t+k|t} &= \left(\frac{P_t^*}{P_{t+k}} \right)^{-\varepsilon} Y_{t+k} \Rightarrow \hat{y}_{t+k} - \hat{y}_{t+k|t} = \varepsilon (\hat{p}_t^* - \hat{p}_{t+k}). \\
 \Rightarrow \widehat{mc}_{t+k|t}^r &= \widehat{mc}_{t+k}^r - \frac{\alpha \varepsilon}{1-\alpha} (\hat{p}_t^* - \hat{p}_{t+k}).
 \end{aligned}$$

With CRS: $\alpha = 0 \Rightarrow \widehat{mc}_{t+k|t}^r = \widehat{mc}_{t+k}^r$. The real marginal cost is common across firms regardless whether they can reset price at t .

Since

$$\widehat{mc}_{t+k|t}^r = \widehat{mc}_{t+k}^r - \frac{\alpha \varepsilon}{1-\alpha} (\hat{p}_t^* - \hat{p}_{t+k}),$$

we have

$$\hat{p}_t^* = (1 - \beta\theta) E_t \sum_{k=0}^{\infty} (\beta\theta)^k [\Theta \widehat{mc}_{t+k}^r + \hat{p}_{t+k}], \quad \Theta = \frac{1-\alpha}{1-\alpha+\alpha\varepsilon} \leq 1.$$

Derivation:

$$\begin{aligned}
 \hat{p}_t^* &= (1 - \beta\theta) \mathbb{E}_t \left\{ \sum_{k=0}^{\infty} (\beta\theta)^k \left[\widehat{mc}_{t+k}^r - \frac{\alpha \varepsilon}{1-\alpha} (\hat{p}_t^* - \hat{p}_{t+k}) + \hat{p}_{t+k} \right] \right\}, \\
 \hat{p}_t^* &= -\frac{\alpha \varepsilon}{1-\alpha} \hat{p}_t^* + (1 - \beta\theta) \mathbb{E}_t \left\{ \sum_{k=0}^{\infty} (\beta\theta)^k \left[\widehat{mc}_{t+k}^r + \frac{1-\alpha+\alpha\varepsilon}{1-\alpha} \hat{p}_{t+k} \right] \right\}, \\
 \frac{1-\alpha+\alpha\varepsilon}{1-\alpha} \hat{p}_t^* &= (1 - \beta\theta) \mathbb{E}_t \left\{ \sum_{k=0}^{\infty} (\beta\theta)^k \left[\widehat{mc}_{t+k}^r + \frac{1-\alpha+\alpha\varepsilon}{1-\alpha} \hat{p}_{t+k} \right] \right\}, \\
 \hat{p}_t^* &= (1 - \beta\theta) \mathbb{E}_t \left\{ \sum_{k=0}^{\infty} (\beta\theta)^k [\Theta \widehat{mc}_{t+k}^r + \hat{p}_{t+k}] \right\}.
 \end{aligned}$$

$$\begin{aligned}
\hat{p}_t^* &= (1 - \beta\theta)(\Theta\widehat{mc}_t^r + \hat{p}_t) + \beta\theta\mathbb{E}_t\hat{p}_{t+1}^* \quad (*), \\
\beta\theta\hat{p}_{t+1}^* &= \beta\theta(1 - \beta\theta)\mathbb{E}_t\left\{\sum_{k=0}^{\infty}(\beta\theta)^k[\Theta\widehat{mc}_{t+1+k}^r + \hat{p}_{t+1+k}]\right\} \\
&= (1 - \beta\theta)\mathbb{E}_t\left\{\sum_{k=0}^{\infty}(\beta\theta)^{k+1}[\Theta\widehat{mc}_{t+(k+1)}^r + \hat{p}_{t+(k+1)}]\right\}.
\end{aligned}$$

Let $k' = k + 1, \forall k$:

$$\beta\theta\hat{p}_{t+1}^* = (1 - \beta\theta)\mathbb{E}_t\left\{\sum_{k'=1}^{\infty}(\beta\theta)^{k'}[\Theta\widehat{mc}_{t+k'}^r + \hat{p}_{t+k'}]\right\}.$$

k' is any index; we can use k . (*) just take out the term of $k = 0$. Combine (*) with log-linearized price index:

$$\begin{aligned}
\hat{p}_t &= \theta\hat{p}_{t-1} + (1 - \theta)\hat{p}_t^*, \\
\hat{p}_t &= \theta\hat{p}_{t-1} + (1 - \theta)(1 - \beta\theta)(\Theta\widehat{mc}_t^r + \hat{p}_t) + (1 - \theta)\beta\theta\mathbb{E}_t\hat{p}_{t+1}^*, \\
\text{since } \hat{p}_{t+1}^* &= \frac{\mathbb{E}_t\hat{p}_{t+1} - \theta\hat{p}_t}{1 - \theta}, \\
\hat{p}_t &= \theta\hat{p}_{t-1} + (1 - \theta)(1 - \beta\theta)(\Theta\widehat{mc}_t^r + \hat{p}_t) + (1 - \theta)\beta\theta\frac{\mathbb{E}_t\hat{p}_{t+1} - \theta\hat{p}_t}{1 - \theta}.
\end{aligned}$$

$$\begin{aligned}
\hat{p}_t &= \theta\hat{p}_{t-1} + (1 - \theta)(1 - \beta\theta)\Theta\widehat{mc}_t^r + (1 - \theta - \beta\theta + \beta\theta^2)\hat{p}_t + \beta\theta\mathbb{E}_t\hat{p}_{t+1} - \beta\theta^2\hat{p}_t, \\
\hat{p}_t - \theta\hat{p}_{t-1} - (1 - \theta)\hat{p}_t &= (1 - \theta)(1 - \beta\theta)\Theta\widehat{mc}_t^r + \beta\theta(\mathbb{E}_t\hat{p}_{t+1} - \hat{p}_t), \\
\theta(\hat{p}_t - \hat{p}_{t-1}) &= (1 - \theta)(1 - \beta\theta)\Theta\widehat{mc}_t^r + \beta\theta\mathbb{E}_t\hat{\pi}_{t+1}, \\
\hat{\pi}_t &= \underbrace{\frac{(1 - \theta)(1 - \beta\theta)}{\theta}}_{\lambda}\Theta\widehat{mc}_t^r + \beta\mathbb{E}_t\hat{\pi}_{t+1}.
\end{aligned}$$

$$\Rightarrow \hat{\pi}_t = \beta\mathbb{E}_t\hat{\pi}_{t+1} + \lambda\widehat{mc}_t^r \quad \text{iterate forward}$$

$$\Rightarrow \hat{\pi}_t = \lambda\sum_{k=0}^{\infty}\beta^k\mathbb{E}_t\widehat{mc}_{t+k}^r.$$

λ is the slope of NKPC w.r.t. real marginal cost.

$\theta \uparrow$ sticky price $\Rightarrow \lambda \downarrow \Rightarrow$ flatter Phillips curve.

$\Theta \downarrow$ stronger decreasing returns $\Rightarrow \lambda \downarrow$.

Inflation results from aggregate consequences of purposeful price-setting decisions by firms, which adjust their prices in light of current and anticipated cost conditions.

Relate marginal cost with output gap

$$\begin{aligned}\widehat{mc}_t^r &= \hat{w}_t - \hat{p}_t - \frac{1}{1-\alpha}(\hat{a}_t - \alpha\hat{y}_t), \\ \hat{w}_t - \hat{p}_t &= \sigma\hat{c}_t + \varphi\hat{n}_t \Rightarrow \hat{w}_t - \hat{p}_t = \sigma\hat{y}_t + \varphi\hat{n}_t, \\ \Rightarrow \widehat{mc}_t^r &= \frac{\varphi + \alpha + \sigma - \alpha\sigma}{1-\alpha}\hat{y}_t - \frac{\varphi + 1}{1-\alpha}\hat{a}_t.\end{aligned}$$

Real marginal cost rises with output, falls with productivity.

1. Labor becomes harder to expand due to household requires more compensation.
2. Decreasing return to scale: labor becomes less productive at the margin.

Under flexible price:

$$\begin{aligned}P_t(i) &= \frac{\varepsilon}{\varepsilon - 1} \frac{W_t}{A_t F_{N,t}(N_t(i))} = \mu MC_t = P_t, \\ MC_t^r &= \frac{MC_t}{P_t} = \frac{1}{\mu} \Rightarrow \widehat{mc}_t^r = 0, \quad \text{real marginal cost has no variation.}\end{aligned}$$

Define the output under flexible price as natural output (long-run output) Y_t^n .

$$\Rightarrow \text{log deviation } \hat{y}_t^n \Rightarrow \hat{y}_t^n = \frac{\varphi + 1}{\varphi + \alpha + \sigma - \alpha\sigma} \hat{a}_t.$$

Define output gap as $\tilde{y}_t = \hat{y}_t - \hat{y}_t^n$.

$$\begin{aligned}\text{Then } \widehat{mc}_t^r &= \frac{\varphi + \alpha + \sigma - \alpha\sigma}{1-\alpha}(\hat{y}_t^n + \tilde{y}_t) - \frac{\varphi + 1}{1-\alpha}\hat{a}_t \\ &= \frac{\varphi + \alpha + \sigma - \alpha\sigma}{1-\alpha} \frac{\varphi + 1}{\varphi + \alpha + \sigma - \alpha\sigma} \hat{a}_t + \frac{\varphi + \alpha + \sigma - \alpha\sigma}{1-\alpha} \tilde{y}_t - \frac{\varphi + 1}{1-\alpha} \hat{a}_t \\ &= \frac{\varphi + \alpha + \sigma(1-\alpha)}{1-\alpha} \tilde{y}_t = \left(\sigma + \frac{\varphi + \alpha}{1-\alpha} \right) \tilde{y}_t.\end{aligned}$$

NKPC:

$$\begin{aligned}\hat{\pi}_t &= \beta \mathbb{E}_t \hat{\pi}_{t+1} + \lambda \widehat{mc}_t^r \quad (\hat{\pi}_{t+1} = \pi_{t+1}) \\ \Rightarrow \hat{\pi}_t &= \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa \tilde{y}_t, \quad \kappa = \lambda \left(\sigma + \frac{\varphi + \alpha}{1-\alpha} \right), \quad \lambda = \frac{(1-\theta)(1-\beta\theta)}{\theta} \Theta, \quad \Theta = \frac{1-\alpha}{1-\alpha + \alpha\varepsilon}.\end{aligned}$$

When demand pushes output above natural output, firms face tighter resource constraints, and the marginal cost rises.

1. σ : demand / consumption smoothing channel.

$$Y_t > Y_t^n \Rightarrow C_t > C_t^n \Rightarrow MU \text{ of leisure } \uparrow \Rightarrow \text{tends to work less, but to keep } N \Rightarrow \frac{W}{P} \uparrow \Rightarrow MC^r \uparrow.$$

$\sigma \uparrow \Rightarrow$ less willing to substitute consumption intertemporally
 $\Rightarrow$ stronger intertemporal C, N substitution
 $\Rightarrow$ stronger wage pressure $\Rightarrow$ stronger MC pressure.

2. φ : labor market channel.

$\varphi \uparrow \Rightarrow$ households dislike work $\Rightarrow W$ increases by more $\Rightarrow MC^r$ increases by more.

3. α : diminishing-returns-to-scale channel.

$\alpha \uparrow \Rightarrow$ to increase output by 1 unit $\Rightarrow \alpha \uparrow$ units increase in labor $\Rightarrow W/P \uparrow \Rightarrow MC^r \uparrow$.

Dynamic IS curve

$$\begin{aligned}\hat{y}_t &= \mathbb{E}_t\{\hat{y}_{t+1}\} - \frac{1}{\sigma} \left(\hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\} \right), \\ \hat{y}_t^n + \tilde{y}_t &= \mathbb{E}_t\{\hat{y}_{t+1}^n + \tilde{y}_{t+1}\} - \frac{1}{\sigma} \left(\hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\} \right), \\ \tilde{y}_t &= \mathbb{E}_t\{\hat{y}_{t+1}^n - \hat{y}_t^n\} + \mathbb{E}_t\{\tilde{y}_{t+1}\} - \frac{1}{\sigma} (\hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\}).\end{aligned}$$

Define natural interest rate as:

$$\begin{aligned}\hat{r}_t^n &= \sigma \mathbb{E}_t\{\hat{y}_{t+1}^n - \hat{y}_t^n\} = \frac{\sigma(1+\varphi)}{\varphi + \alpha + \sigma(1-\alpha)} \mathbb{E}_t\{\Delta \hat{a}_{t+1}\} = \sigma \psi_{ya}^n \mathbb{E}_t\{\Delta \hat{a}_{t+1}\}. \\ \Rightarrow \tilde{y}_t &= -\frac{1}{\sigma} (\hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^n) + \mathbb{E}_t\{\tilde{y}_{t+1}\}.\end{aligned}$$

Define

$$\begin{aligned}\hat{r}_t &= \hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\}. \\ \Rightarrow \tilde{y}_t &= -\frac{1}{\sigma} (\hat{r}_t - \hat{r}_t^n) + \mathbb{E}_t\{\tilde{y}_{t+1}\}.\end{aligned}$$

Assume $\lim_{T \rightarrow \infty} \mathbb{E}_t\{\tilde{y}_{t+T}\} = 0$ and iterate forward:

$$\Rightarrow \tilde{y}_t = -\frac{1}{\sigma} \mathbb{E}_t \left\{ \sum_{k=0}^{\infty} (\hat{r}_{t+k} - \hat{r}_{t+k}^n) \right\}.$$

Output gap is proportional to the sum of current and anticipated deviations between the real interest rate and natural real interest rate.

NKPC:

$$\begin{aligned}\hat{\pi}_t &= \beta \mathbb{E}_t\{\hat{\pi}_{t+1}\} + \kappa \tilde{y}_t \quad \text{iterate forward} \\ \Rightarrow \hat{\pi}_t &= \kappa \sum_{k=0}^{\infty} \beta^k \mathbb{E}_t\{\tilde{y}_{t+k}\}, \quad \text{where } \kappa = \lambda \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right).\end{aligned}$$

Inflation today is the discounted sum of the expected future output gap.

Basic New Keynesian non-policy block:

$$\begin{cases} \hat{\pi}_t = \kappa \sum_{k=0}^{\infty} \beta^k \mathbb{E}_t\{\tilde{y}_{t+k}\}, & \text{NKPC: determines inflation given output gap path,} \\ \tilde{y}_t = -\frac{1}{\sigma} \mathbb{E}_t \left\{ \sum_{k=0}^{\infty} (\hat{r}_{t+k} - \hat{r}_{t+k}^n) \right\}, & \text{DIS: determines output gap given real interest rate gap path.} \end{cases}$$

To close, we need more equations to pin down how nominal interest rate i / $\hat{i}_t$ evolves (monetary policy).

$$\begin{cases} \tilde{y}_t = -\frac{1}{\sigma} \left(\hat{i}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^n \right) + \mathbb{E}_t\{\tilde{y}_{t+1}\} & \dashrightarrow \text{DIS (Demand)} \\ \hat{\pi}_t = \beta \mathbb{E}_t\{\hat{\pi}_{t+1}\} + \kappa \tilde{y}_t & \dashrightarrow \text{NKPC (Supply)} \end{cases}$$

Dynamic monetary policy rule

$$\hat{i}_t = \phi_\pi \hat{\pi}_t + \phi_y \tilde{y}_t + v_t, \quad v_t \text{ is exogenous with zero mean.}$$

$$\phi_\pi > 0, \quad \phi_y > 0, \quad \text{assume S.S. } \bar{\Pi} = 0.$$

DIS:

$$\begin{aligned} \tilde{y}_t &= -\frac{1}{\sigma} [\phi_\pi \hat{\pi}_t + \phi_y \tilde{y}_t - \mathbb{E}_t\{\hat{\pi}_{t+1}\} - (\hat{r}_t^n - v_t)] + \mathbb{E}_t\{\tilde{y}_{t+1}\} \quad (***) \\ \Rightarrow \left(1 + \frac{\phi_y}{\sigma}\right) \tilde{y}_t + \frac{\phi_\pi}{\sigma} \hat{\pi}_t &= \mathbb{E}_t\{\tilde{y}_{t+1}\} + \frac{1}{\sigma} \mathbb{E}_t\{\hat{\pi}_{t+1}\} + \frac{1}{\sigma} (\hat{r}_t^n - v_t). \\ -\kappa \tilde{y}_t + \hat{\pi}_t &= \beta \mathbb{E}_t\{\hat{\pi}_{t+1}\}. \end{aligned}$$

Write in matrix form in system of $[\tilde{y}_t, \hat{\pi}_t]'$:

$$\begin{aligned} \underbrace{\begin{bmatrix} 1 + \phi_y/\sigma & \phi_\pi/\sigma \\ -\kappa & 1 \end{bmatrix}}_M \begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} &= \begin{bmatrix} 1 & 1/\sigma \\ 0 & \beta \end{bmatrix} \begin{bmatrix} \mathbb{E}_t\{\tilde{y}_{t+1}\} \\ \mathbb{E}_t\{\hat{\pi}_{t+1}\} \end{bmatrix} + \begin{bmatrix} 1/\sigma \\ 0 \end{bmatrix} (\hat{r}_t^n - v_t). \\ \begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} &= M^{-1} \begin{bmatrix} 1 & 1/\sigma \\ 0 & \beta \end{bmatrix} \begin{bmatrix} \mathbb{E}_t\{\tilde{y}_{t+1}\} \\ \mathbb{E}_t\{\hat{\pi}_{t+1}\} \end{bmatrix} + M^{-1} \begin{bmatrix} 1/\sigma \\ 0 \end{bmatrix} (\hat{r}_t^n - v_t). \\ M^{-1} &= \frac{1}{|M|} \begin{bmatrix} 1 & -\phi_\pi/\sigma \\ \kappa & 1 + \phi_y/\sigma \end{bmatrix} = \Omega \begin{bmatrix} \sigma & -\phi_\pi \\ \kappa\sigma & \sigma + \phi_y \end{bmatrix}, \quad \Omega = \frac{1}{\sigma + \phi_y + \kappa\phi_\pi}. \\ A_T &= \Omega \begin{bmatrix} \sigma & -\phi_\pi \\ \kappa\sigma & \sigma + \phi_y \end{bmatrix} \begin{bmatrix} 1 & 1/\sigma \\ 0 & \beta \end{bmatrix} = \begin{bmatrix} \sigma & 1 - \beta\phi_\pi \\ \kappa\sigma & \kappa + \beta(\sigma + \phi_y) \end{bmatrix} \Omega, \\ B_T &= \Omega \begin{bmatrix} \sigma & -\phi_\pi \\ \kappa\sigma & \sigma + \phi_y \end{bmatrix} \begin{bmatrix} 1/\sigma \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ \kappa \end{bmatrix} \Omega. \\ \Rightarrow \begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} &= A_T \begin{bmatrix} \mathbb{E}_t\{\tilde{y}_{t+1}\} \\ \mathbb{E}_t\{\hat{\pi}_{t+1}\} \end{bmatrix} + B_T (\hat{r}_t^n - v_t). \end{aligned}$$

Both $\tilde{y}_t$ and $\hat{\pi}_t$ are not predetermined $\xRightarrow{BK}$ both eigenvalues of A_T are within the unit circle.

For a 2×2 matrix A_T , both eigenvalues lie inside unit circle iff Jury/Schur condition holds:

$$\begin{cases} 1 - \det(A_T) > 0 & (D < 1), \\ 1 - \text{tr}(A_T) + \det(A_T) > 0 & (p(1) > 0), \\ 1 + \text{tr}(A_T) + \det(A_T) > 0 & (p(-1) > 0). \end{cases}$$

$$\det(A_T) = \Omega^2 [\sigma(\kappa + \beta(\sigma + \phi_y)) - (1 - \beta\phi_\pi)\kappa\sigma] = \Omega^2 \sigma \beta (\sigma + \phi_y + \kappa\phi_\pi) = \frac{\beta\sigma}{\sigma + \phi_y + \kappa\phi_\pi},$$

$$\text{tr}(A_T) = \Omega [\sigma + \kappa + \beta(\sigma + \phi_y)] = \frac{\sigma + \kappa + \beta(\sigma + \phi_y)}{\sigma + \phi_y + \kappa\phi_\pi}.$$

$$1^\circ \quad 1 - \det(A_T) = 1 - \frac{\beta\sigma}{\sigma + \phi_y + \kappa\phi_\pi} = \frac{(1 - \beta)\sigma + \phi_y + \kappa\phi_\pi}{\sigma + \phi_y + \kappa\phi_\pi} > 0 \quad \text{satisfy,}$$

$$\beta < 1, \sigma > 0, \phi_y > 0, \kappa > 0, \phi_\pi > 0.$$

$$\begin{aligned} 2^\circ \quad 1 - \text{tr}(A_T) + \det(A_T) &= 1 - \frac{\sigma + \kappa + \beta(\sigma + \phi_y)}{\sigma + \phi_y + \kappa\phi_\pi} + \frac{\beta\sigma}{\sigma + \phi_y + \kappa\phi_\pi} \\ &= \frac{\kappa(\phi_\pi - 1) + (1 - \beta)\phi_y}{\sigma + \phi_y + \kappa\phi_\pi} > 0 \end{aligned}$$

$$\Rightarrow \kappa(\phi_\pi - 1) + (1 - \beta)\phi_y > 0 \quad \text{satisfy if.}$$

$$3^\circ \quad 1 + \text{tr}(A_T) + \det(A_T) = \frac{(\sigma + \phi_y + \kappa\phi_\pi) + (\sigma + \kappa + \beta(\sigma + \phi_y)) + \beta\sigma}{\sigma + \phi_y + \kappa\phi_\pi} > 0 \quad \text{satisfy.}$$

$$\text{The system is stable iff:} \quad \kappa(\phi_\pi - 1) + (1 - \beta)\phi_y > 0.$$

If $\phi_y = 0 \Rightarrow \phi_\pi > 1$, consistent with the Taylor principle: when inflation rises by 1%, the central bank needs to raise the nominal interest rate by $> 1\%$.

$$r_t = i_t - \mathbb{E}_t \pi_{t+1} \quad \text{Fisher equation.}$$

$$\mathbb{E}_t \pi_{t+1} \uparrow 1\%, \quad i_t \uparrow > 1\% \Rightarrow r_t \uparrow \Rightarrow \mathbb{E}_t \pi_{t+1} \downarrow.$$

Monetary policy shock

Assume v_t follows AR(1):

$$v_t = \rho_v v_{t-1} + \varepsilon_t^v, \quad \rho_v \in [0, 1].$$

$\varepsilon_t^v > 0$: contractionary MP shock, $\varepsilon_t^v < 0$: expansionary MP shock.

For simplicity, shut down technology shock:

$$\hat{a}_t = 0 \Rightarrow \hat{y}_t^n = 0, \quad \hat{r}_t^n = 0, \quad \forall t.$$

Guess

$$\tilde{y}_t = \psi_{yv} v_t, \quad \hat{\pi}_t = \psi_{\pi v} v_t.$$

Plug into (***):

$$\begin{aligned} \psi_{yv} v_t &= -\frac{1}{\sigma} (\phi_\pi \psi_{\pi v} v_t + \phi_y \psi_{yv} v_t + v_t - \rho_v \psi_{\pi v} v_t) + \rho_v \psi_{yv} v_t, \\ -\sigma \psi_{yv} &= \phi_\pi \psi_{\pi v} + \phi_y \psi_{yv} + 1 - \rho_v \psi_{\pi v} - \sigma \rho_v \psi_{yv}, \\ [\sigma(1 - \rho_v) + \phi_y] \psi_{yv} &= (\rho_v - \phi_\pi) \psi_{\pi v} - 1 \quad (1). \end{aligned}$$

From NKPC:

$$\begin{aligned} \hat{\pi}_t &= \beta \mathbb{E}_t \{ \hat{\pi}_{t+1} \} + \kappa \tilde{y}_t, \\ \psi_{\pi v} v_t &= \beta \rho_v \psi_{\pi v} v_t + \kappa \psi_{yv} v_t, \\ \psi_{yv} &= \frac{1 - \beta \rho_v}{\kappa} \psi_{\pi v} \quad (2). \end{aligned}$$

Plug (2) into (1):

$$\psi_{\pi v} = -\frac{\kappa}{(1 - \beta \rho_v)[\sigma(1 - \rho_v) + \phi_y] + \kappa(\phi_\pi - \rho_v)}.$$

$$\hat{\pi}_t = \underbrace{-\kappa \Lambda_v}_{<0} v_t$$

$$\Lambda_v = \frac{1}{(1 - \beta \rho_v)[\sigma(1 - \rho_v) + \phi_y] + \kappa(\phi_\pi - \rho_v)} > 0.$$

where

$$\kappa = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \lambda = \left(\sigma + \frac{\varphi + \alpha}{1 - \alpha} \right) \frac{(1 - \theta)(1 - \beta\theta)}{\theta} \frac{1 - \alpha}{1 - \alpha + \alpha\varepsilon} > 0.$$

$$\tilde{y}_t = \underbrace{-(1 - \beta \rho_v) \Lambda_v}_{<0} v_t$$

$$\begin{aligned} \hat{r}_t &= \hat{i}_t - \mathbb{E}_t \{ \hat{\pi}_{t+1} \} = \phi_\pi \psi_{\pi v} v_t + \phi_y \psi_{yv} v_t + v_t + \kappa \Lambda_v \rho_v v_t \\ &= \underbrace{\sigma(1 - \rho_v)(1 - \beta \rho_v) \Lambda_v}_{>0} v_t. \end{aligned}$$

$$\hat{i}_t = \underbrace{[\sigma(1 - \rho_v)(1 - \beta\rho_v) - \kappa\rho_v]}_{\text{coefficient on } v_t} \Lambda_v v_t.$$

ambiguous; depends on ρ_v . if $\rho_v \uparrow$, $\hat{i}_t \downarrow$.

If monetary policy is more persistent, $\hat{i}_t \downarrow$ real interest rate decreases.

$$\text{Therefore } \begin{cases} v_t = \rho_v v_{t-1} + \varepsilon_t^v, & \rho_v \in [0, 1], \\ \hat{\pi}_t = -\kappa \Lambda_v v_t, \\ \tilde{y}_t = -(1 - \beta\rho_v) \Lambda_v v_t, \\ \hat{r}_t = \sigma(1 - \rho_v)(1 - \beta\rho_v) \Lambda_v v_t, \\ \hat{i}_t = [\sigma(1 - \rho_v)(1 - \beta\rho_v) - \kappa\rho_v] \Lambda_v v_t. \end{cases}$$

Calibration: $\beta = 0.99$ (quarterly data), $\sigma = 1$ (log utility), $\varphi = 1$, $\alpha = 1/3$, $\varepsilon = 6$, $\theta = 2/3$, $\bar{y} = 4$.

With $\phi_\pi = 1.5$, $\phi_y = 0.5$, $\rho_v = 0.5$, and shock $\varepsilon_t^v = 25\text{bp}$: $\hat{\pi}_t < 0$, $\tilde{y}_t < 0$, $\hat{y}_t < 0$, $\hat{i}_t > 0$, $\hat{m}_t < 0$.

Technology shock

Assume an AR(1) process of $\hat{a}_t$:

$$\hat{a}_t = \rho_a \hat{a}_{t-1} + \varepsilon_t^a, \quad \rho_a \in [0, 1].$$

$$\hat{r}_t^n = \sigma \mathbb{E}_t \{ \hat{y}_{t+1}^n - \hat{y}_t^n \} = \sigma \psi_{ya}^n \mathbb{E}_t \{ \Delta \hat{a}_{t+1} \} = -\sigma \psi_{ya}^n (1 - \rho_a) \hat{a}_t.$$

Setting $v_t = 0, \forall t$. Assume:

$$\begin{aligned} \tilde{y}_t &= \psi_{yr} \hat{r}_t^n, & \hat{\pi}_t &= \psi_{\pi r} \hat{r}_t^n. \\ \begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \end{bmatrix} &= A_T \begin{bmatrix} \mathbb{E}_t \{ \tilde{y}_{t+1} \} \\ \mathbb{E}_t \{ \hat{\pi}_{t+1} \} \end{bmatrix} + B_T (\hat{r}_t^n - v_t). \end{aligned}$$

Opposite sign to v_t :

$$\begin{aligned} \psi_{yr} &= -\psi_{yv}, & \psi_{\pi r} &= -\psi_{\pi v}. \\ \tilde{y}_t &= (1 - \beta\rho_a) \Lambda_a \hat{r}_t^n = -\sigma \psi_{ya}^n (1 - \rho_a) (1 - \beta\rho_a) \Lambda_a \hat{a}_t, \\ \hat{\pi}_t &= \kappa \Lambda_a \hat{r}_t^n = -\sigma \psi_{ya}^n (1 - \rho_a) \kappa \Lambda_a \hat{a}_t, \end{aligned}$$

where

$$\Lambda_a = \frac{1}{(1 - \beta\rho_a)[\sigma(1 - \rho_a) + \phi_y] + \kappa(\phi_\pi - \rho_a)} > 0.$$

$$\begin{aligned} \hat{y}_t &= \hat{y}_t^n + \tilde{y}_t = \psi_{ya}^n \hat{a}_t - \sigma \psi_{ya}^n (1 - \rho_a) (1 - \beta\rho_a) \Lambda_a \hat{a}_t \\ &= \psi_{ya}^n \left[1 - \underbrace{\sigma(1 - \rho_a)(1 - \beta\rho_a) \Lambda_a}_{(*)} \right] \hat{a}_t. \end{aligned}$$

$$\begin{aligned}
1 - (*) &= 1 - \sigma(1 - \rho_a)(1 - \beta\rho_a) \frac{1}{\sigma(1 - \rho_a)(1 - \beta\rho_a) + \phi_y(1 - \beta\rho_a) + \kappa(\phi_\pi - \rho_a)} \\
&= \frac{\sigma(1 - \rho_a)(1 - \beta\rho_a) + \phi_y(1 - \beta\rho_a) + \kappa(\phi_\pi - \rho_a) - \sigma(1 - \rho_a)(1 - \beta\rho_a)}{\sigma(1 - \rho_a)(1 - \beta\rho_a) + \phi_y(1 - \beta\rho_a) + \kappa(\phi_\pi - \rho_a)} \\
&= \frac{\phi_y(1 - \beta\rho_a) + \kappa(\phi_\pi - \rho_a)}{\sigma(1 - \rho_a)(1 - \beta\rho_a) + \phi_y(1 - \beta\rho_a) + \kappa(\phi_\pi - \rho_a)} \\
&\Rightarrow \hat{y}_t = \frac{\phi_y(1 - \beta\rho_a) + \kappa(\phi_\pi - \rho_a)}{\sigma(1 - \rho_a)(1 - \beta\rho_a) + \phi_y(1 - \beta\rho_a) + \kappa(\phi_\pi - \rho_a)} \psi_{ya}^n \hat{a}_t.
\end{aligned}$$

$$\begin{aligned}
\phi_y(1 - \beta\rho_a) + \kappa(\phi_\pi - \rho_a) &= \kappa(\phi_\pi - 1) + (1 - \beta)\phi_y + \kappa - \rho_a\kappa + \beta\phi_y - \beta\rho_a\phi_y \\
&= \underbrace{\kappa(\phi_\pi - 1) + (1 - \beta)\phi_y}_{\text{stable condition}} + \underbrace{(1 - \rho_a)\kappa}_{0 \leq \rho_a < 1} + \underbrace{(1 - \rho_a)\beta\phi_y}_{0 \leq \rho_a < 1} > 0.
\end{aligned}$$

y_t moves positively with a_t .

$$(1 - \alpha)\hat{n}_t = \hat{y}_t - \hat{a}_t = [(\psi_{ya}^n - 1) - \sigma\psi_{ya}^n(1 - \rho_a)(1 - \beta\rho_a)\Lambda_a] \hat{a}_t.$$

In calibration: $\sigma = 1 \Rightarrow \psi_{ya}^n = \frac{1+\varphi}{\varphi+\alpha+\sigma(1-\alpha)} = 1 \Rightarrow n_t$ declines when a_t increases.

Equilibrium with exogenous money supply: $\Delta\hat{m}_t$

$$\frac{M_t}{P_t} = L(i_t, Y_t).$$

Close the model with money demand equation instead of Taylor rule. Define log real money balance:

$$\hat{\mathbb{M}}_t = \hat{m}_t - \hat{p}_t \Rightarrow \hat{\mathbb{M}}_t = \hat{y}_t - \eta\hat{i}_t.$$

$$\Rightarrow \tilde{y}_t - \eta\hat{i}_t = \hat{\mathbb{M}}_t - \hat{y}_t^n \Rightarrow \hat{i}_t = \frac{1}{\eta} \left(\tilde{y}_t + \hat{y}_t^n - \hat{\mathbb{M}}_t \right).$$

Output $\uparrow \Rightarrow$ money demand $\uparrow \Rightarrow i \uparrow$. Real money supply $\uparrow \Rightarrow i \downarrow$.

$$DIS \Rightarrow (1 + \sigma\eta)\tilde{y}_t = \sigma\eta\mathbb{E}_t\{\tilde{y}_{t+1}\} + \hat{\mathbb{M}}_t + \eta\mathbb{E}_t\{\hat{\pi}_{t+1}\} + \eta\hat{r}_t^n - \hat{y}_t^n.$$

From $\hat{\mathbb{M}}_t$ definition:

$$\hat{\mathbb{M}}_t - \hat{\mathbb{M}}_{t-1} = \Delta\hat{m}_t - \hat{\pi}_t \Rightarrow \hat{\mathbb{M}}_{t-1} = \hat{\mathbb{M}}_t + \hat{\pi}_t - \Delta\hat{m}_t.$$

$$A_{M,0} \begin{bmatrix} \tilde{y}_t \\ \hat{\pi}_t \\ \hat{\mathbb{M}}_{t-1} \end{bmatrix} = A_{M,1} \begin{bmatrix} \mathbb{E}_t\{\tilde{y}_{t+1}\} \\ \mathbb{E}_t\{\hat{\pi}_{t+1}\} \\ \hat{\mathbb{M}}_t \end{bmatrix} + B_M \begin{bmatrix} \hat{r}_t^n \\ \hat{y}_t^n \\ \Delta\hat{m}_t \end{bmatrix}.$$

$\widehat{M}_{t-1}$ is predetermined and $\tilde{y}_t, \hat{\pi}_t$ are not predetermined. Stationary solution will exist iff

$$A_M = A_{M,0}^{-1} A_{M,1} \text{ has two eigenvalues inside the unit circle.}$$

Assume the monetary policy shock follows AR(1):

$$\Delta \hat{m}_t = \rho_m \Delta \hat{m}_{t-1} + \varepsilon_t^m, \quad \rho_m \in [0, 1].$$

Let

$$\hat{r}_t^n = \hat{y}_t^n = 0, \quad \varepsilon_t^m = 0.25, \quad t = 0, \quad \varepsilon_t^m = 0, \quad \forall t \neq 0.$$

An expansionary monetary policy shock:

$$\Delta \hat{m}_t \uparrow \Rightarrow \tilde{y}_t \uparrow, \hat{\pi}_t \uparrow, \widehat{M}_t \uparrow, \quad \hat{i}_t \uparrow: \text{ no liquidity effect } (\sigma = 1), \quad \hat{r}_t \downarrow.$$

$$\hat{a}_t \uparrow \Rightarrow \hat{r}_t^n \downarrow, \hat{y}_t \uparrow, \hat{\pi}_t \downarrow, \hat{n}_t \downarrow, \hat{r}_t \uparrow, \hat{i}_t \approx 0, \quad \hat{y}_t^n \uparrow \uparrow, \tilde{y}_t \downarrow.$$

Liquidity effect: when real money balance $\uparrow$, $\widehat{M}_t \uparrow \Rightarrow$ households hold too much money than they want.

$$\Rightarrow \hat{i}_t \downarrow \text{ to induce households to hold that money.}$$

Why $\sigma = 1$ liquidity effect is gone?

$$\hat{i}_t = \frac{1}{\eta} \left(\tilde{y}_t + \hat{y}_t^n - \widehat{M}_t \right) \quad \text{with } \hat{y}_t^n = 0.$$

DIS:

$$\tilde{y}_t = -\frac{1}{\sigma} (\hat{i}_t - \mathbb{E}_t \hat{\pi}_{t+1}) + \mathbb{E}_t \tilde{y}_{t+1}.$$

Set $\hat{y}_t^n = 0$:

$$\hat{i}_t = \frac{1}{\eta} (\tilde{y}_t - \widehat{M}_t).$$

Liquidity effect: $\Delta \hat{m}_t > 0, \hat{i}_t < 0$, requires $\widehat{M}_t > \tilde{y}_t$. Real balance must rise by more than the output gap.

$$DIS \Rightarrow \hat{i}_t = \underbrace{\mathbb{E}_t \hat{\pi}_{t+1}}_{>0} + \sigma \underbrace{[\mathbb{E}_t \tilde{y}_{t+1} - \tilde{y}_t]}_{<0}.$$

$$\hat{i}_t < 0 \Leftrightarrow \sigma (\tilde{y}_t - \mathbb{E}_t \tilde{y}_{t+1}) > \mathbb{E}_t \hat{\pi}_{t+1}.$$

After expansionary money shock, $\tilde{y}_t \uparrow \Rightarrow \tilde{y}_t > \mathbb{E}_t \tilde{y}_{t+1} \Rightarrow \mathbb{E}_t \tilde{y}_{t+1} - \tilde{y}_t < 0$.

When $\sigma = 1$, the negative term is not large enough to dominate the positive term.

Liquidity effect disappears whenever:

$$\mathbb{E}_t \hat{\pi}_{t+1} > \tilde{y}_t - \mathbb{E}_t \tilde{y}_{t+1}.$$

Thus, $\sigma \gg 1, \rho_m \ll 1$ liquidity effect presents; money shock not persistent $\Rightarrow$ expected inflation $\downarrow$.